HIGH-LEVEL DESIGN (HLD)

1. System Overview

Problem Definition

Traditional customer support systems and chatbots fail to provide accurate answers because they do not use real-time document-based knowledge. They often generate incorrect or generic responses.

Solution

A Retrieval-Augmented Generation (RAG)-based Customer Support Assistant that:

- Processes a PDF knowledge base
- Retrieves relevant information using embeddings
- Generates contextual answers using an LLM
- Uses workflow orchestration
- Supports Human-in-the-Loop (HITL) escalation

Scope

- Single PDF-based system
- CLI-based interaction
- Context-aware responses
- Basic escalation system

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2. Architecture Diagram

User → Interface → Workflow Engine → Retriever → LLM → Response / HITL

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3. Components Description

Document Loader

Loads PDF using PyPDFLoader and extracts text.

Chunking Strategy

Splits document into chunks (500 size, 50 overlap) to preserve context.

Embedding Model

Uses HuggingFace embeddings to convert text into vector form.

Vector Store

Uses ChromaDB to store embeddings for efficient retrieval.

Retriever

Fetches top-k relevant chunks based on similarity.

LLM

Generates answers using retrieved context.

Workflow Engine

Uses LangGraph to manage execution flow.

Routing Layer

Decides whether to answer or escalate.

HITL Module

Escalates queries to human when system confidence is low.

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4. Data Flow

PDF → Text → Chunks → Embeddings → Vector DB
User Query → Retriever → Context → LLM → Output

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5. Technology Choices

- ChromaDB → Lightweight, fast vector search
- LangGraph → Graph-based workflow control
- HuggingFace → Free and accessible LLM

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6. Scalability Considerations

- Supports large documents via chunking
- Can scale to multiple documents
- Retrieval optimized for performance
- Can handle increased query load